W?

TEAM NGỌA HỔ TÀNG LONG

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What-If AI

Simulate before you decide — Validate with data — Track realized profit.

38.8 ms
101 prices × 5,000 draws

18 / 19
Beat naive baseline
SYNTHETIC backtest

0 Digits
Calculated by LLM

Controlled Econometrics + Go Monte Carlo

Pricing · Restock · Cash Runway · Journal



Prototype v0.1 + MVP v0.2 · Local-First Decision Workspace

Live Verified

Retail SMEs Price Blindly Because Spreadsheets Look Back and Generic AI Guesses.

WHY THIS EXISTS

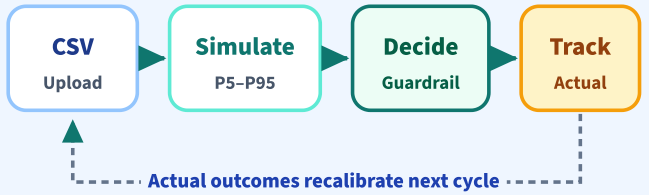
- **Gut-feel moves:** Prices, promos & restocks rely on intuition or backward sheets.
- **Messy POS CSVs:** VN headers, dot-thousands & missing costs stall analysis.
- **Peak trap:** 11.11 / Tết surges mimic elasticity: **−5.9** naive vs **−1.5** true SYNTHETIC.
- **Unsafe AI chat:** Generic chatbots fabricate cash figures.

Target: Vietnamese Retail SMEs (20–300 SKUs)

WHAT IT IS

"One CSV in, one decision out."

Offline laptop workspace in **VI | EN** turning sales exports into profit ranges.



Runs Offline **Zero API Key Required**

CORE VALUE

- **Simulate before you decide:** See safe-to-upside profit bands (P5–P95).
- **Validate with data:** Safety elasticity cutoff, holdout proof & drift alerts.
- **Track realized profit:** Compare plan vs actual store cash outcomes.
- **Zero LLM math:** Fail-closed numeric checks protect owners.

Decision Transparency First

SHOP OWNER
"Will cash stay above my buffer?"

MERCHANDISER
"Which price grows profit safely?"

MARKETING
"Did our promo pay off net?"

INVENTORY OPS
"When and how much to reorder?"

Sources: README.md · docs/VALIDATION.md (hard_event_confounded_12)

Excel Sees Yesterday and Chatbots Guess — Only What-If AI Pairs Simulation With Provenance.

Decision Capability	Excel / Static BI	Generic LLM Chat	What-If AI
Forward Risk Simulation	Single fixed formula	Unverified prose estimate	5,000 draws P5–P95 range
Zero Fabricated Numbers	Manual sheet audit	Hallucination risk	Fail-closed check discards mismatch
VN Holiday Confounding	Manual cleanup	Uncontrolled guesses	Calendar GLM SYNTHETIC −5.9 → −1.8
Target-Price Search	Slow 1-cell Goal Seek	Cannot optimize constraints	38.8 ms across 101 constrained prices
Raw CSV Privacy	Local workbook	Prompts expose raw rows	RAM-only CSV · ≤5 masked fields

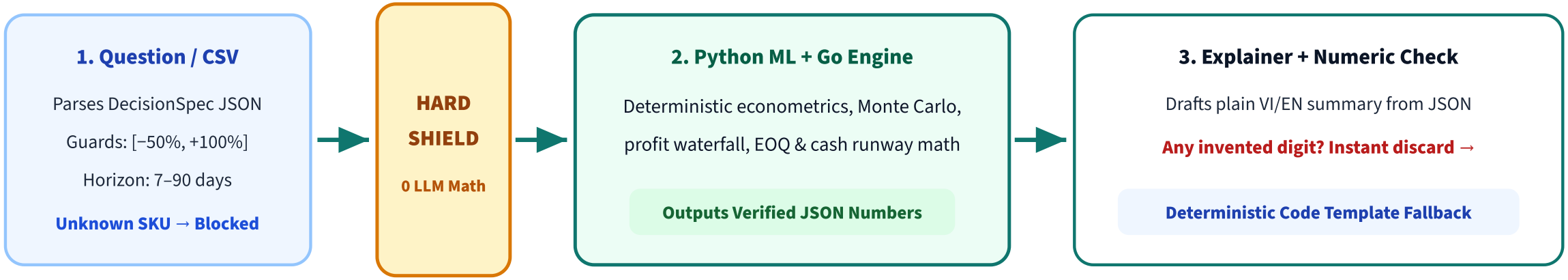
1. HISTORICAL INPUTS
Observed sales, price & unit cost

2. MODEL ESTIMATES
Controlled buyer reaction + CI range

3. USER ASSUMPTIONS
Promo days, budget cap & cash floor

Mandatory Provenance Badges Across UI & Deck: YOUR DATA · SYNTHETIC · PUBLIC (M5) · PUBLIC (DUNNHUMBY)

A Hard Shield Keeps Language AI Out of Every Calculation Step.



OFFLINE COLUMN MAPPING

33 / 33 (100%)

Messy CSV headers mapped at **≥ 0.55** confidence offline.

NLQ SAFETY GUARDRAILS

30/30 · 15/15

30/30 valid specs parsed · **15/15** unsafe prompts blocked.

ZERO CLOUD LOCK-IN

No API Key Needed

Rule-based parser & template advisor run fully offline.

Methodology footnote: Column matching uses Hungarian assignment; NLQ parses against strict Pydantic bounds and runs `verify_explanation_numbers()` before rendering.

Three Clean Tiers Separate UI Presentation, Fast Simulation, and Statistical Storage.



TIER 1 · PRESENTATION ONLY

- **Zero math in browser:** Renders engine JSON & formats VND display.
- **Privacy guarantee:** Never persists raw sales rows in `localStorage`.

Static Vite + Nginx Container

TIER 2 · GO 1.22+ (GO1.27.3 TESTED)

- Parallel price optimization, Monte Carlo & cash planning kernels.
- Unified gateway strips duplicate upstream CORS headers.

Deterministic Parallel Kernel

TIER 3 · PYTHON 3.10+ (DOCKER 3.12-SLIM)

- POS CSV normalization, seasonality control & holdout backtesting.
- Drift alarms, causal lift & persistent local `decisions.db`.

FastAPI 0.110+ · Statsmodels

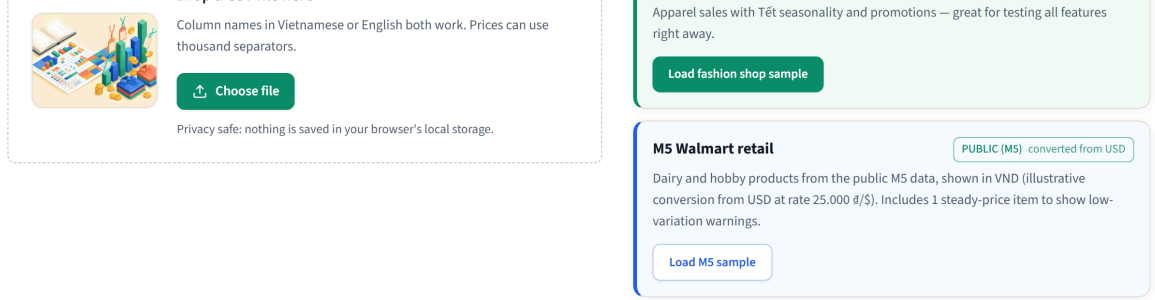
0.0057% – 0.0219%
Go vs Python P50 parity (< 0.5% gate)

GOMAXPROCS 1 = 8+
Bit-identical seeded JSON output

docker compose up
Single-command laptop startup

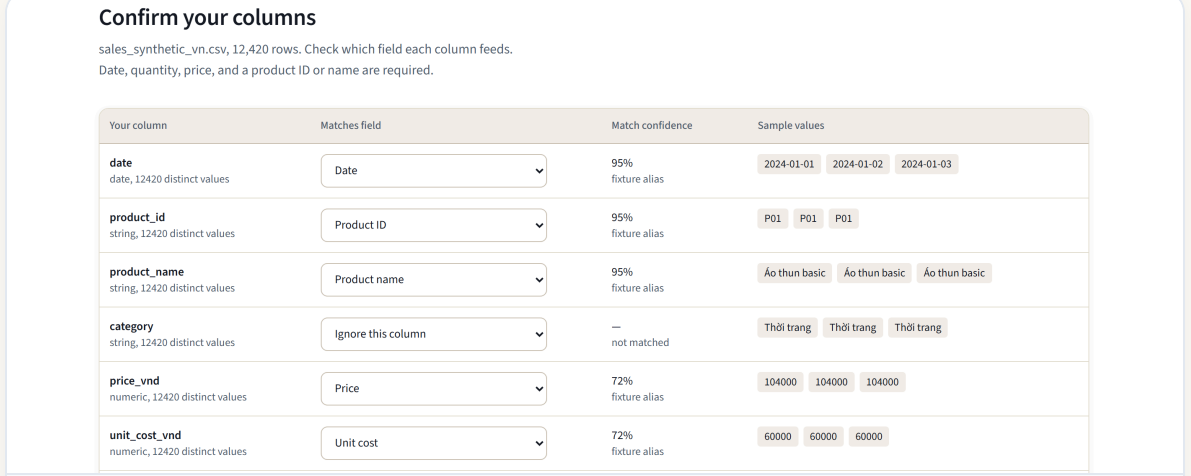
Methodology footnote: Parallel Monte Carlo draws stay deterministic across thread counts via counter-based SplitMix64 indexed per simulation run.

One Consistent Flow Turns Raw Sales CSVs Into Recommended Prices in Four Clear Screens.



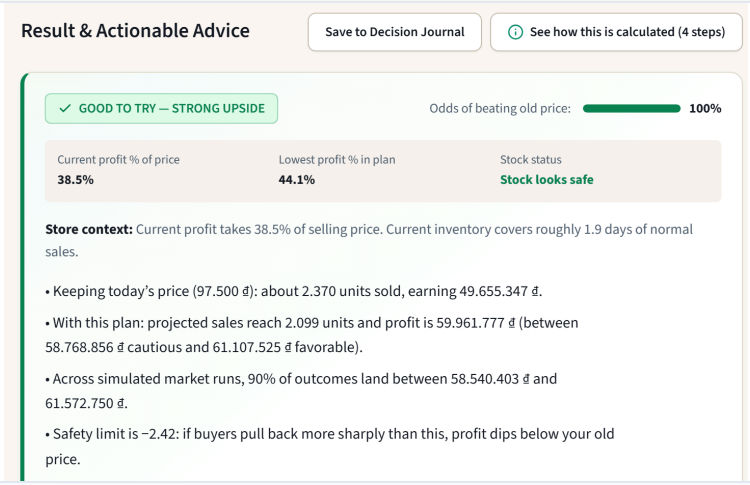
1. Drop messy CSV or choose sample

No Browser Storage



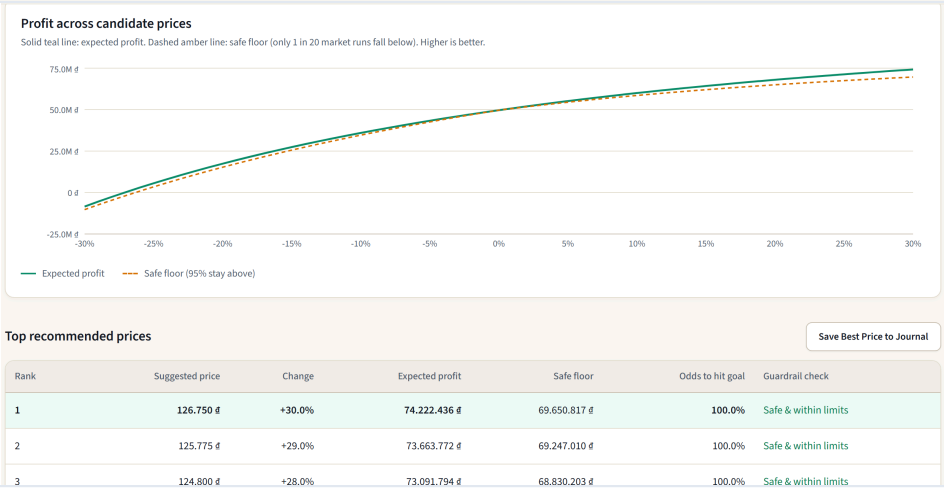
2. Auto-match 12,420 POS rows with confidence

95% / 72% Match



3. P01 +10%: clear verdict, P5–P95 range & −2.42 cutoff

100% Odds



4. Reverse search ranks safe candidate prices

Rank #1 Verified

Zoomed UI crops from What-If AI live React build (runs offline or backed by Go + Python APIs)

Deterministic Engine Formulas Match Exact Analytical Reference Math Digit-for-Digit.

BASELINE SETUP (0% PRICE MOVE)

Price P0 = 100.000 ₫ · Cost C = 60.000 ₫ · Qty Q0 = 10.000

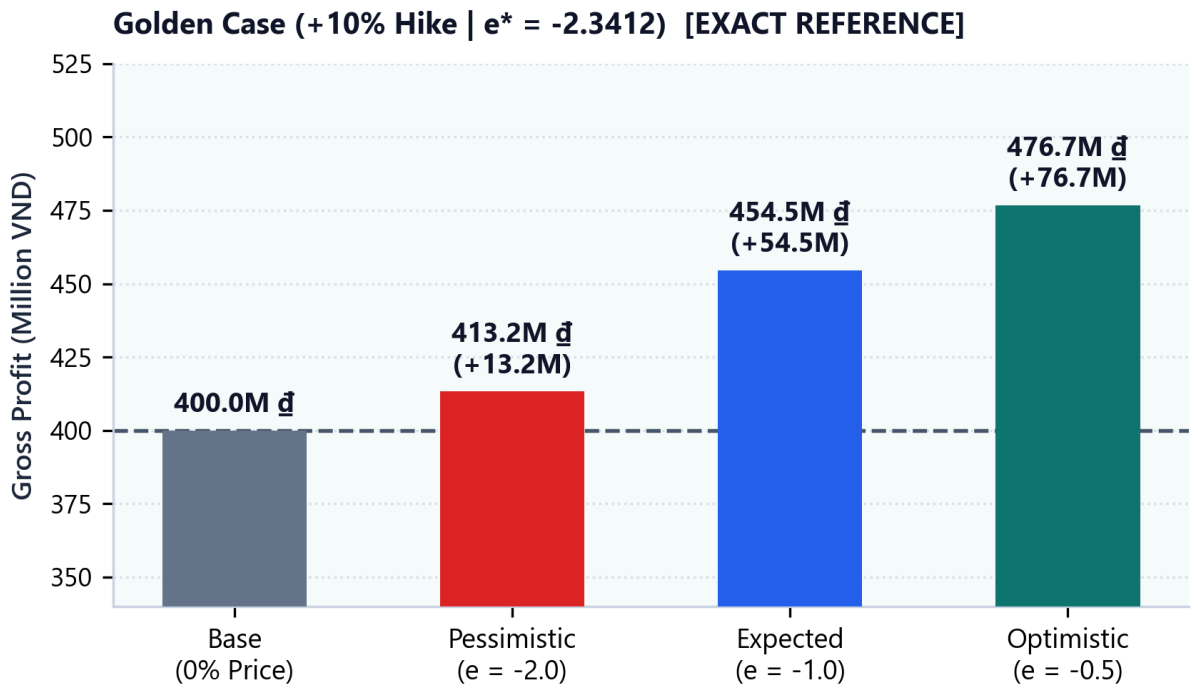
Base Profit = **400,0M ₫** · Test Hike: **+10% (110.000 ₫)**

Scenario	Reaction e	Gross Profit
Pessimistic	−2.0	**413,2M ₫** (413.223.141)
Expected	−1.0	**454,5M ₫** (454.545.455)
Optimistic	−0.5	**476,7M ₫** (476.731.295)

SAFETY BREAK-EVEN ELASTICITY

e* = −2.3412

Exact match with reference formula

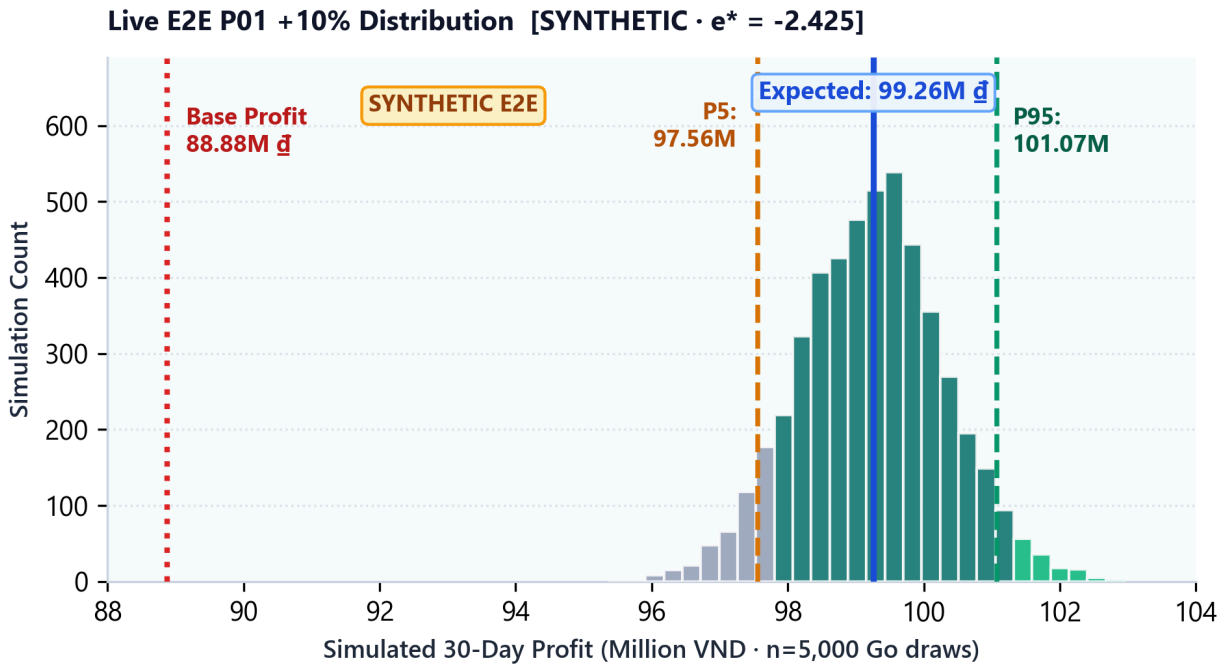


Verified in Go & Python golden unit suites

Zero Floating Drift

Sources: README.md · engine-go/internal/sim/simulate_test.go · e2e/cross_check_py_go.py

Simulate Profit Tails, Launch Multi-Phase Playbooks, or Search Target Prices in 38.8 ms.



P01 +10% (30d): Base 88,88M → P5 97,56M · Exp 99,26M · P95 101,07M ₫ SYNTHETIC · e* = −2.425

6 PRE-BUILT PLAYBOOK TEMPLATES

Editable Business Assumptions

1. 7d Flash Sale
2. Tết Clearance
3. Marketing Push
4. Stock Expansion
5. Cash Preservation
6. New SKU Ramp

REVERSE TARGET-PRICE FINDER

101 Prices × 5,000 Draws in 38.8 ms

- Ranks highest expected or safest floor price automatically.
- Enforces minimum margin & stock cap constraints.
- Live E2E P01 +10% passes with SYNTHETIC e* = −2.425 safety margin.

Profit Goal | Safe Floor P5 | Zero Constraint Violations

Sources: e2e/demo_path.py (P01 +10% live run) · docs/ENGINE.md (38.8 ms) · web/src/data/scenarioTemplates.ts

Keep Cash Above Your Safety Buffer While Balancing Stockout Risk Across SKUs.

1. Reorder Quantity & Timing

Compute exact batch sizes and reorder alerts under capital limits.

2. Cash Safety Protection

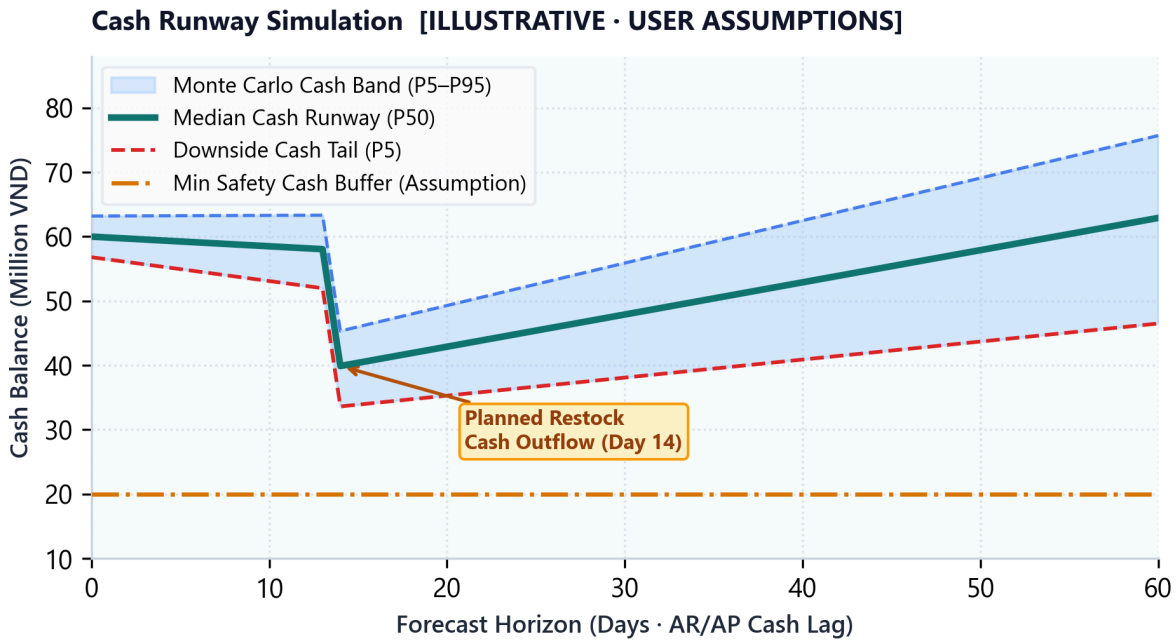
Model payment delays so cash never dips below your safety buffer.

3. Multi-SKU Capital Split

Allocate tight cash & shelf space toward risk-adjusted profit.

4. Test Before Committing

Decide whether to test a price on a small slice before scaling storewide.



60-day cash trajectory under user supplier/payment lag assumptions

ILLUSTRATIVE · USER ASSUMPTIONS

Methodology footnote: EOQ + safety stock (Acklam inverse-normal quantile z) · Cash Conversion Cycle (DIO + DSO − DPO) · 250-bucket DP knapsack with P5 downside penalty · Bayesian EVPI / EVSI (200 × 40 trials).

Know When Predictions Drift, Measure Real Promo Lift, and Audit Every Past Call.

Know When Predictions Drift

Flags shifting shopper behavior, seasonal turning points & rising forecast errors before bad margins compound.

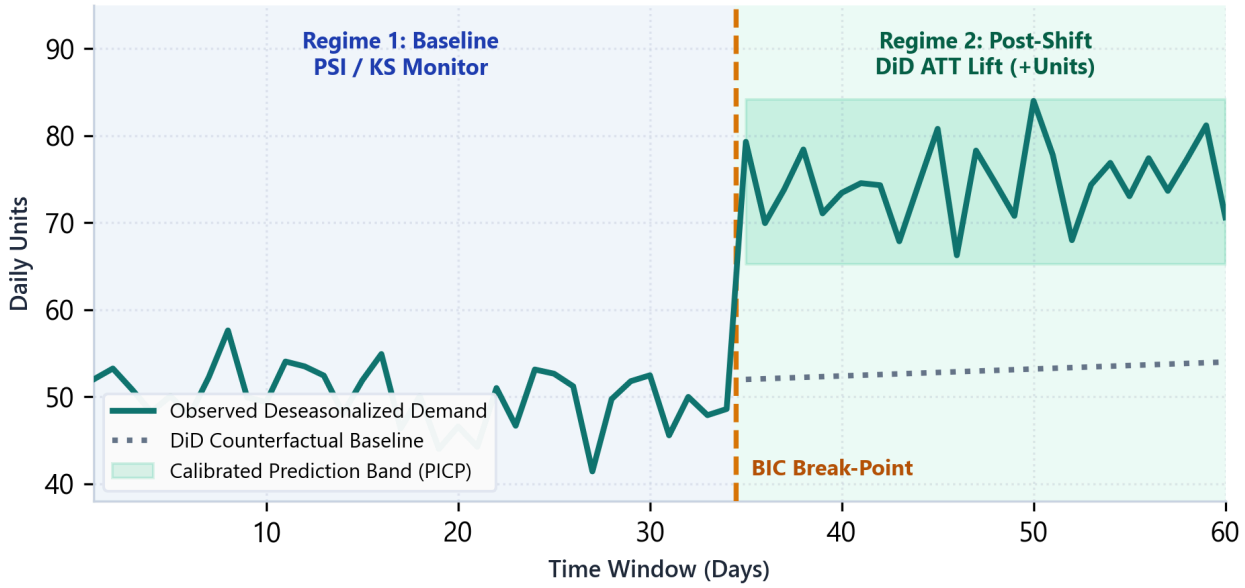
Audit True Promo Impact

Compares promoted SKUs against steady control items to report real incremental profit, not raw volume.

Track Realized Profit Over Time

Saves every simulated range and logs actual post-launch profit to verify prediction reliability.

Regime Detection (BIC), PICP Calibration & DiD Causal Audit [METHOD DIAGRAM]



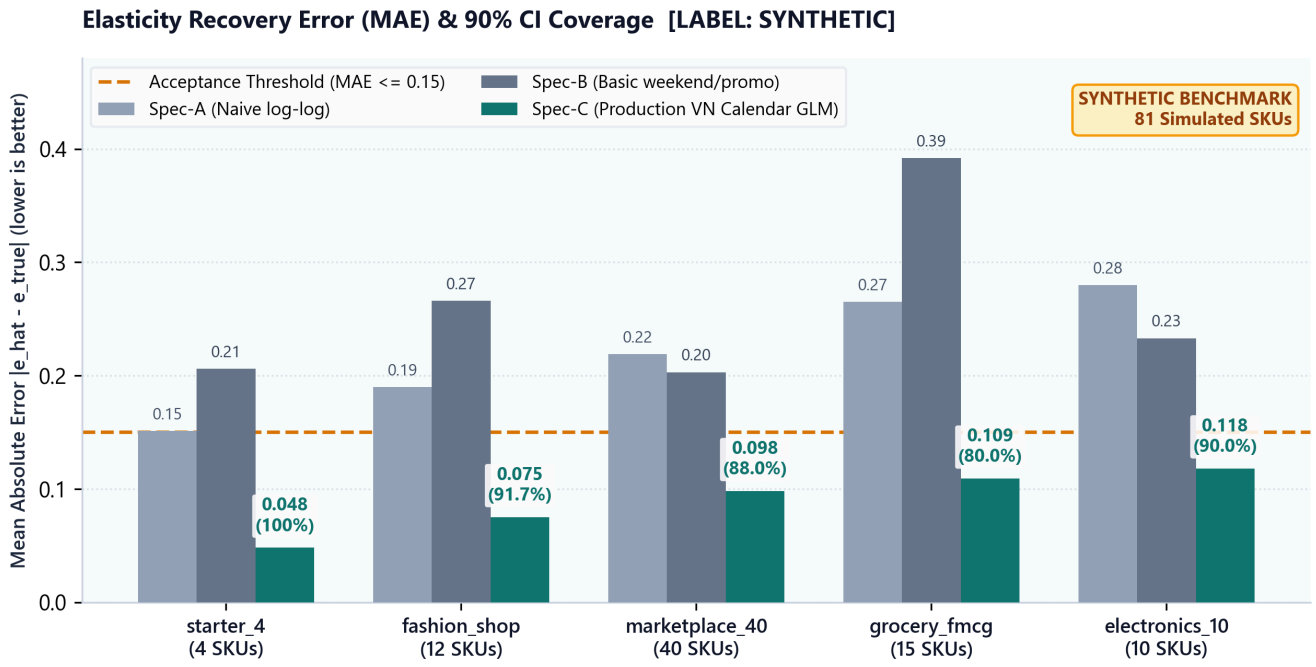
Demand shift alerts + forecast bands + control baseline METHOD DIAGRAM

4-STEP DECISION FEEDBACK LOOP (LOCAL SQLITE)

1. Decision → 2. Actual Outcome → 3. Audit vs P5–95 → 4. Update SQLite

Methodology footnote: Rolling PICP@50/80/90, MPIW, Pinball Loss & Bracher WIS · BIC binary-segmentation regimes · PSI (8 bins) / KS / Durbin-Watson drift · HAC DiD (maxlags 2–3) & Segmented ITS · SQLite `decisions.db`.

Calendar-Controlled Modeling Cuts Elasticity Error Below 0.12 and Blocks Flat-Price Guesses.



5 stores (81 SKUs): Spec-C cuts error 2–3× vs naive

SYNTHETIC

STANDARD 5-STORE ACCURACY

SYNTHETIC

- **MAE 0.048 – 0.118** across 81 SKUs (target ≤ 0.15).
- **90% CI covers 80.0% – 100.0%**; Spearman rank **0.791 – 1.000**.

11.11 / TẾT CONFOUNDING STRESS

SYNTHETIC

Reduces naive distortion **−5.9** down to **−1.8** vs true **−1.5** (12 hard SKUs).

HONEST FLAT-PRICE GUARD

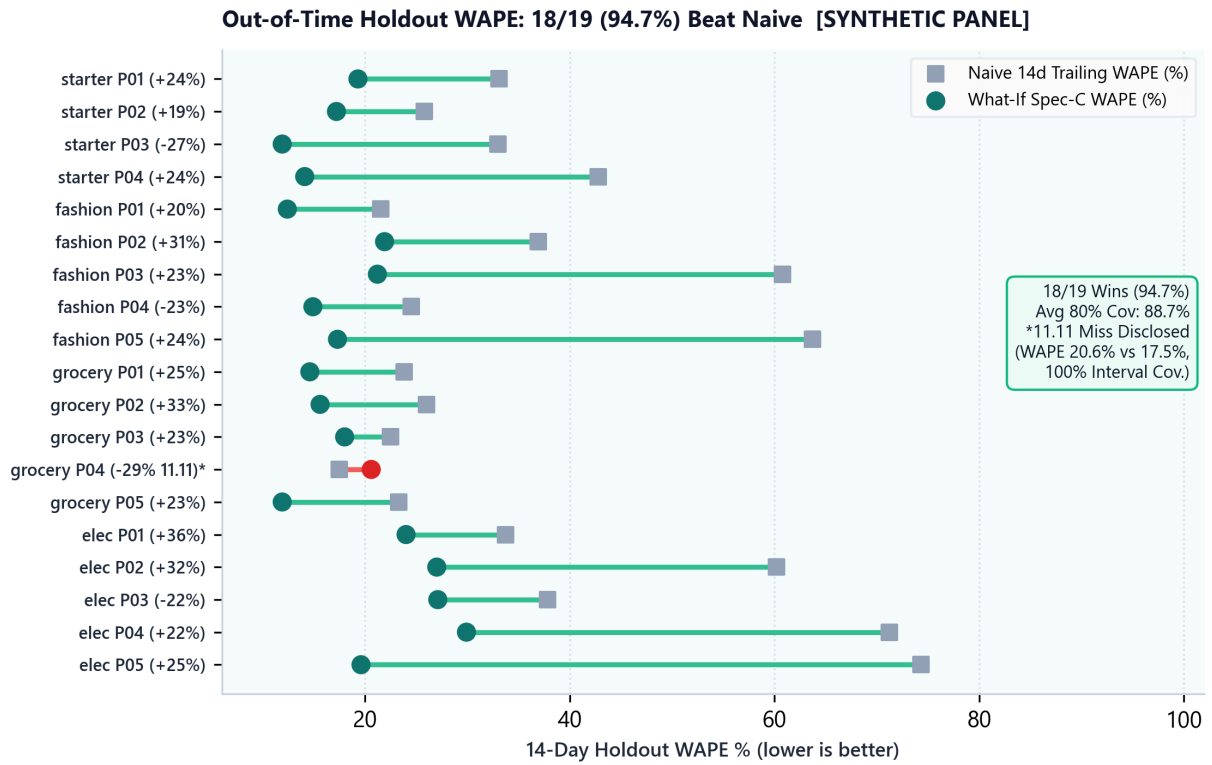
SYNTHETIC

8 / 8 (100%) steady-price SKUs flagged instead of guessing certainty.

Disclosure: Ground-truth sensitivity tests run on synthetic stores — never presented as live VN receipts.

Methodology footnote: Spec-C Poisson GLM controls Tết peaks, double-day events (11.11/12.12), paydays & promos; flat-price guard fires when `sd(ln P) < 0.03` (test fixture sd=0.013).

18 of 19 Historical Holdouts Beat Baseline — Including Open Disclosure of Our 11.11 Miss.



Zero future leakage · t < split → 14d forecast

SYNTHETIC · 18/19 (94.7%)

HOLDOUT ACCURACY

SYNTHETIC

18 / 19 (94.7%) · 88.7% Cov

Beats trailing 14-day baseline; 80% uncertainty band captures **88.7%** of real holdout days.

HONEST FAILURE DISCLOSURE

SYNTHETIC

grocery_fmcg_15 P04 (11.11, −28.7%): model WAPE **20.6%** vs naive **17.5%** (100% band coverage). Saturation cap shifted response mid-promo.

PUBLIC METHOD CHECK & AUTOMATED SUITES

PUBLIC

M5 & Dunnhumby verify noisy external ingestion (25.000 ₫/$ illustrative). Verified today: **39 Pytest · 23 Go (-race) · 17 Vitest · 8/8 E2E · 3/3 Cross-Check**.

Sources: docs/VALIDATION.md · live pytest/go test/vitest/E2E runs (13 Nov 2026)

Role-Ready Decision Clarity With Zero Lock-In and Local-Only CSV Privacy.

SHOP OWNER

Protects cash buffer before PO commitments.

MERCHANDISER

Ranks safe prices in **38.8 ms**.

MARKETING

Audits true promo lift beyond hype.

INVENTORY OPS

Balances reorder batch vs stockout risk.

SAFER, FASTER DECISIONS

- Replaces slow backward spreadsheets with **P5–P95** risk bands.
- Blocks flat-price illusions via strict variation guards.
- Audits realized outcomes vs plan to reduce repeat mistakes.

Zero LLM Hallucinated Figures

LAPTOP & PRIVACY DEPLOYMENT

- Reads POS CSV exports directly — zero ERP lock-in.
- Runs completely offline on laptop (`docker compose up`).
- Raw CSV rows never leave RAM, hit `localStorage`, or enter LLM prompts.

Local Memory Only · Bilingual VI | EN

PILOT KPIS & COMMERCIAL ROLLOUT

Weekly prep turnaround, stockout prevention, P5–P95 hit rate & gross margin stability — **to be measured in the 60-day pilot**; packaging validated with pilot partners.

Zero Upfront Software Lock-In

Clear Model Boundaries Protect Owners Today While Guiding Schema v2.

CURRENT GUARDED BOUNDARIES (DOCS/LIMITS.MD)

- **Controlled observational estimate:** Not A/B price-test certainty; unobserved shocks remain possible.
- **Single-SKU scope today:** Cross-product cannibalization waits for basket `order_id` history.
- **No competitor reaction model:** Automated rivalry awaits `competitor_price_vnd` feeds.
- **Hard safety boundaries:** Flags flat prices (`sd(ln P) < 0.03`), zero-stock censoring & enforces **[−50%, +100%] / 7–90d**.

Flags Uncertainty Instead of Guessing

SCHEMA V2 & POST-HACKATHON ROADMAP

- **Phase A — Basket Bundles:** Ingest `order_id` receipts for cross-elasticity & Buy-X-Get-Y bundles.
- **Phase B — Rival Price Tracking:** Ingest `competitor_price_vnd` for live price-gap simulation.
- **Phase C — Read-Only POS Adapters:** Optional scheduled connectors while keeping CSV upload instant.

Only Ships Models Supported by Store Data

MVP CSV SCOPE TODAY

`date · sku · price · qty · unit_cost · promo · stock`

Schema v2 Upgrade →

NEXT FIELDS UNLOCKED

`order_id` (bundles) · `competitor_price_vnd` (rivals)

Sources: docs/LIMITS.md · docs/DECISION_INTELLIGENCE_AUDIT.md · ml-py/app/elasticity/fit.py

Next Step: Validate Real-World Profit Lift With 3 Vietnamese Retail SMEs Over 60 Days.

WEEKS 1–2 · ONBOARDING

CSV Setup & Calibration

Clean store exports, verify margins & flag flat-price SKUs before touching live prices.

WEEKS 3–6 · SHADOW RUNS

Simulate Before Action

Run weekly price, promo & restock plans alongside ops; log calls to local SQLite.

WEEKS 7–8 · REALIZED AUDIT

Outcome Proof & Sign-Off

Measure realized profit vs P5–P95 bands, stockout avoidance & prep time saved.

OUR ASK FOR PILOT PARTNERS

- **3 Vietnamese Retail SMEs** (20–300 SKUs, fashion / FMCG / home).
- **≥ 90 days POS CSV exports** with unit cost & promo calendar dates.
- **30 minutes / week** owner check-in to record realized outcomes.

Gate: Exit Pilot if Realized Accuracy Misses Expectations

TEAM NGỌA HỔ TÀNG LONG

Built End-to-End with Zero LLM Math

• **Nguyễn Hoàng Tùng** — Team Leader
• **Phạm Viết Trường** — Core Member
• **Ngô Thanh Tùng** — Core Member

Thank you · Ready for Q&A

What-If AI · 2026